



IT Training Report

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Position: IT Trainee

Company: Northern Electricity Distribution Company (شركة توزيع كهرباء الشمال)

Duration: September 2025 – November 2025 (3 Months)

Location: Nablus, Palestine

Introduction

This report presents a comprehensive summary of my **three-month Information Technology (IT) training** at the Northern Electricity Distribution Company in Nablus, Palestine. The training combined theoretical knowledge with practical experience in IT infrastructure, operating systems, enterprise systems, databases, networking, and cybersecurity, providing hands-on exposure to real industrial environments and large-scale technical systems.

First Month (September) – Orientation, Core IT Infrastructure & Operating Systems

The first month focused on orientation, understanding the company's technical environment, and building a strong foundation in IT infrastructure and operating systems.

Key Activities:

- **Department Familiarization:** Introduction to all technical departments and their coordination workflows supporting electricity distribution services.
- **Server Infrastructure:** Visited server rooms to observe data handling, backup procedures, and the TABS data management system.
- **Technical Support:** Assisted in troubleshooting hardware and software issues, device maintenance, and user support.
- **Printing Systems:** Learned IP-based printer installation, driver configuration, maintenance, and differences between modern and legacy printers.
- **Operating Systems:** Installed and troubleshoot Windows operating systems, handled boot issues, drivers, and recovery procedures.
- **Career Development:** Attended sessions on career planning, teamwork, and problem-solving skills for IT professionals.

Key Outcomes:

- Solid understanding of IT department operations.
- Practical experience in OS installation and printer networking.
- Improved troubleshooting and professional communication skills.

Second Month (October) – Enterprise Systems, Databases & Security Systems

The second month emphasized enterprise software systems, databases, and institutional security technologies.

Key Activities:

- **ERB System Training:** Learned to use the Employee Resource and Billing (ERB) system for managing employee data, attendance, payroll, and performance.
- **Report Generation:** Created attendance and financial reports to support management decision-making.
- **Database Fundamentals:** Observed how databases connect departments and ensure data consistency.
- **Billing Systems Overview:** Studied the customer billing lifecycle, including electricity usage data, invoice generation, and payment tracking.
- **Fire Alarm Systems:** Learned about fire detection sensors, control panels, system integration, and emergency protocols.
- **Surveillance Systems:** Gained hands-on experience with CCTV setup, camera types, placement strategies, and system maintenance.
- **Security Policies:** Applied post-installation system security configurations and access controls.

Key Outcomes:

- Proficiency in enterprise systems and report analysis.
- Understanding of database integration and data accuracy.
- Awareness of institutional safety and security systems.

Third Month (November) – Networking, Hardware, Cybersecurity & System Integration

The final month focused on advanced technical skills related to networking, hardware maintenance, and cybersecurity.

Key Activities:

- **Networking Fundamentals:** Studied IP addressing (IPv4/IPv6), MAC addressing, DHCP, static vs dynamic IPs, and routing concepts.
- **Hardware Maintenance:** Assembled and upgraded computers, installed RAM and SSDs, and replaced faulty components.
- **Performance Optimization:** Compared NVMe and SATA SSDs and analyzed system bottlenecks.
- **Database Management:** Practiced data validation, error detection, database merging, and cleaning to maintain data integrity.
- **Encryption & Security:** Learned basic encryption concepts such as public/private key cryptography.
- **Firewalls & VPNs:** Managed firewall rules and configured VPN connections for secure remote access.
- **Remote Desktop:** Set up remote access, user permissions, and monitoring for IT support tasks.

Key Outcomes:

- Strong foundation in networking and system security.
- Hands-on experience with hardware optimization.
- Practical understanding of secure remote access and enterprise IT environments.